

Case Study 1: AI-Based Threat Vector

An Analysis of AI-Generated Phishing Email Campaigns Targeting Filipino Users

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Introduction

AI in the Philippines has revolutionized the nature of online deception. Rather than the clumsily-worded emails that previously typified phishing attacks, attackers are now creating grammatically correct, personalized and super convincing emails that look like banks, government agencies and employers. In this paper, I explore the case study of an AI-powered phishing email campaign targeting the Filipino population, as an example of AI-driven deception. It explains how generative AI can help lower technical barriers to AI-generated fake content, how criminals are using it in conjunction with social engineering, and how it is a new threat to the nation's cybersecurity.

In recent years, the danger of AI deception has been increasing, as fraudsters are merging two key problems into a single solution which can be easily scaled to target high numbers of people. Generative language models can generate thousands of different phishing messages in the time a human writer used to take to write one; each message can be customized to a target by a bit of information collected from a social media platform or a data breach (Schmitt & Flechais, 2023). This analysis focuses on the following: to describe the scope and development of this threat in the context of the Philippines; to examine the technological and psychological support for this threat; and to recommend specific technical and educational countermeasures, grounded on current research and Philippine legislation.

Concepts and Foundations

The use of generative AI is increasingly automated tasks in the cybercrime realm which were once performed by crafty professionals. Now that large language models are able to write fluently within the context of what they are writing, it is less likely that the recipient will be able

to identify a scam using awkward expressions or poor grammar (Schmitt & Flechais, 2023). Such models may be used by a single attacker to craft phishing messages, translate them into local languages, and even have real-time conversations with the victims, all of which will enhance the phishing capabilities.

AI can work in tandem with traditional phishing/impersonation methods in three overlapping ways. The first is to make the content come alive, so the e-mail is as real as possible, the logo is reproduced, and, in the more elaborate scenarios, a synthetic voice or picture is as close to the real thing as possible. Secondly, it facilitates accurate targeting because AI can browse and process a vast amount of publicly shared information, including names, employers, and even recent activity, and tailor a message to the target's name. Third, it enables that automation, so it's possible for one person to send thousands of personalized messages without much manual labor (Schmitt & Flechais, 2023). In the Philippines, the same three pillars are evident as Blancaflor et al. (2024) concluded from their survey of Filipino respondents that they feel they know a lot about Cyber Security, but they still find it challenging to recognize fraudulent e-mails and domains that look like their trusted brands.

As with all phishing scams, whether subjected to AI-driven or manual attacks, the scam's core is the combination of technology and human susceptibility. Technical defenses do not close this window of opportunity as they do not normally take into account the weaknesses of human nature, according to Longtchi et al. (2024), which is why social engineering works. The problem becomes more and more severe with generative AI because it can tweak its "twist" to win over the individual target in real-time, so that it is the human element, not the perimeter of the network, that is the weak point. The concept is closely related to the case discussed in this paper

as the Filipino victims were not fooled from a technical fault of their devices, but through the messages that they could not differentiate from the originals, because they were so convincing.

Analysis and Evaluation

Describe and identify threats of a case.

Usually, a phishing email targeting Filipinos is disguised as an email from a bank, a government agency such as the Bureau of Internal Revenue (BIR) or the Social Security System (SSS) or an e-commerce website or employer. The emails notify recipients of a serious issue with their account, an unclaimed refund, and/or the failure to submit required documentation, and include links to a fraudulent, similar site or attachment. The primary targets are individual bank customers, online shoppers, remote employees or workers of organizations that handle financial transactions, whose credentials are compromised and which can result in exposure of the company's network. A vulnerability was directly documented by Blancaflor et al. (2024) who reported that the Filipino respondents were highly susceptible to phishing emails and homoglyph domains which closely resembled the legitimate one after conducting tests on them.

Typically, there are several phases to the attack. The initial method of an attack is to gather information regarding a victim organization or individual from social media, leaked databases or public records. They then use generative AI to enter a message into the web page, often in the same style and formatting as a well-known institution, and sometimes with a spoofed domain or homoglyph domain (which is a web address that looks like the real domain at first glance because of the use of similar characters) (Blancaflor et al., 2024). Once the victim clicks on the embedded link, he/she enters his/her log-in information or the payment information on the fake page and the attacker uses the data to steal it, sell it or to obtain further access to the victim's employer network.

Technology and Method

In this scenario, the two most relevant generative AI methods are: 1) large language model-based text generation, and 2) adversary-in-the-middle proxy tools that can intercept live logon sessions, which has been a newer technique. One of the advantages of text generation for attackers is that they can generate a more natural, localized and contextually appropriate phishing text, without needing to have native English/Filipino writing skills, which used to be a clue for users to recognize a phishing attack. In more sophisticated campaigns, this text is then integrated with adversary in the middle (AiTM) infrastructure, which includes a stealthy proxy server between the victim and the legitimate site, capturing the password(s) as they are typed in and a successful one-time passcode (1TP) can be stored, transcoded and retrieved (Blancaflor et al., 2025).

The validity of such an attack will depend on the quality of the inputs attackers feed into the AI system. To produce a message that is very similar to the institutions house style, a small sample of what the institution has previously sent out in email communications is usually sufficient for a generative model to produce a very similar message. By substituting a Latin letter with another, but visually similar character of a different alphabet, homoglyph substitution further enhances the realism of a fraudulent domain to make it appear almost identical to the real one (Blancaflor et al., 2024). It's also helpful in deception because it removes the normal safeguards such as spelling errors, awkward phrases and obviously bogus domain names that users have been conditioned to look out for over the years, and forces scrutiny which they have not been conditioned to follow.

Psychological Exploitation

The psychological weapons of urgency and authority are a key part of phishing scams and are used extensively by the AI phishing campaigns. The recipient is now very likely to comply with whatever they are told because they don't want to be hit with a penalty, have their bank account frozen, or face their professions' consequences for failing to do what is said.

Another of the most "exploited psychological factors in social engineering" because of time pressure, says Longtchi et al. (2024), that the victim will not be able to think carefully, but will act impulsively rather than logically.

The combination of this sense of urgency and cues of authority (such as letterheads, official-looking names, formal language and specific reference numbers) allows the attackers to appeal to the victim's instinct to obey a trusted institution instead of relying on their instinct to check if this is true. This is effective because it appeals to a decision making shortcut and not to a technical defect. This is a cognitive limitation and not a knowledge gap, and so it worked with time pressure even with people who were aware of cybersecurity; indeed, Blancaflor et al. (2024) did not find many associations between self-reported awareness and phishing resistance in real phishing exercises.

Case of impact and legal context.

The scams involved in the attacks are likely for the purpose of getting financial account details, a one time passcode, personal identifying details, or outright cash transfers. The impacts to the victim may be anything from depleted bank accounts to loans in their name, to their employers' system being hacked, to years of identity theft. Not only is the loss of money a concern, but victims can also be concerned about their online security and trust in the institutions they will use in the future.

This conduct is clearly covered by the Cybercrime Prevention Act of 2012 (RA 10175) in the Philippines. The intentional acquisition or use of identifying information of another person without right – including the use of a fake website to steal the logon information and personal details – is a form of computer related identity theft under Section 4(b)(3) of Republic Act No. 10175, 2012. A fund transfer using stolen personal information may be considered a computer related transfer as well. AI-generated phishing is a computer-related offense, as well as a violation of the confidentiality of computer data (Republic Act No. 10175, 2012) because the initial credential theft is also a computer-related offense.

Application and Implications

Technical Recommendation

What's the right and secure technical answer to phishing attacks using AI? It's not password and one time codes, it's phishing-resistant multi-factor authentication with hardware security keys or passkeys based on the FIDO2 standard. Even if the adversary in the middle is unable to interrupt the authentication session thanks to the progress of the standard MFA methods (such as SMS codes or push notifications sent through the app), the middle man can still break the authentication session and steal all credentials and codes (Blancaflor et al., 2025). The problem with phishing is that there is a way to circumvent this, which is to have to enter a credential into a site with the same domain as the one the web server belongs to, and if someone enters a credential into a phishing site, it won't log on to the attacker's server anyway.

This is a necessary step as deceptive attacks via AI can be readily conducted by AI, but not as easily by a legitimate site. In a recent study of a large sample of Microsoft accounts, Meyer et al. (2023) found that accounts using strong multi-factor authentication were much less likely to be compromised by the more and more sophisticated attackers than accounts protected

by passwords. As long as the phishing resistant authentication mechanism is not using a cryptographic signature for the domain being phished, phishing resistance will remain unaffected by the improvement of the phishing content with AI.

User Education Recommendation

The non-technical approach involves structured verification training: a method of verifying the request by sending a true or urgent request to a bank using a number, from a physical card, and not a number provided in the suspicious message. This directly challenges the above mentioned urgency and the above mentioned authority tactics because it gives a conscious pause between the emotional trigger and the compliant action, that allows the victim to think instead of being caught up by instinct. But training solely on detection tips did not prove to be sufficient in preventing a successful phishing attack for their Filipino respondents (Blancaflor et al., 2024), and hence there has to be a low effort and easy to follow verification routine that people follow under pressure.

Conclusion

AI created phishing email campaigns geared to Filipino users illustrate how generative AI unites the technical and psychological aspects of cybercrime into a single, highly scalable danger. Using fluent, personalized messages and complementing them with homoglyph domains or adversary in the middle infrastructure, attackers eliminate cues that Filipinos have been accustomed to, and the tactics of urgency and authority take advantage of shortcuts in decision making even those who are aware of such tactics cannot help but succumb to. However, while under the existing law (RA 10175) there are legal remedies to prosecute the fruits of the identity theft and subsequent fraud, this does not preclude the harm from happening in the first place. Phishing resistant authentication and phishing verification training both address the technical

part of the problem (the phishing attack) and the human part of the problem (the susceptibility of the user), and have a good chance of reducing the success rate of AI-generated deception in the Philippines.

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